

Antioxidant and antiinflammatory activities of anthocyanins and their aglycon, cyanidin, from tart cherries.



The anthocyanins (1-3) and cyanidin isolated from tart cherries exhibited in vitro antioxidant and antiinflammatory activities comparable to commercial products. The inhibition of lipid peroxidation of anthocyanins 1-3 and their aglycon, cyanidin, were 39, 70, 75, and 57%, respectively, at 2-mM concentrations. The antioxidant activities of 1-3 and cyanidin were comparable to the antioxidant activities of tert-butylhydroquinone and butylated hydroxytoluene and superior to vitamin E at 2-mM concentrations. In the antiinflammatory assay, cyanidin gave IC₅₀ values of 90 and 60 mM, respectively, for prostaglandin H endoperoxide synthase-1 and prostaglandin H endoperoxide synthase-2 enzymes.